OS-Banana-2

Phase 1:

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$
```

Phase 2

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ xxd .pes/objects/2a/594d39232787fb
a8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf | head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$
```

Phase 3

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ ./pes add file1.txt file2.txt
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)
```

```
  untracked:  README.md

vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ cat .pes/index
100644 a5309708f7804f8207340240d6f9356c2b4dfb81c8fd463660ee863c536fb0d8 17766813
77 9 file1.txt
100644 a4ffc512be57913428f753539fa25707cf58131a818cdbc8daa9b897cdea9187 17766813
77 9 file2.txt
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$
```

Screenshot 3B: cat .pes/index showing the text-format index

Phase 4

```
commit 91520c830f300affb486d333542c5f6c54893673085f6341333fd1957a45ade7
Author: Vedanth
Date:   1776682047

    Update file.txt

commit 0e4755b3e6e04203ff56cf499314753b26b15e9a7c555ae8460b79f7dc7ca149
Author: Vedanth
Date:   1776682047

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
46434aa7034a963aa15496c5d5acbc3271c42d482091b760b784f8f1e2f787b6

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0e/4755b3e6e04203ff56cf499314753b26b15e9a7c555ae8460b79f7dc7ca149
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/46/434aa7034a963aa15496c5d5acbc3271c42d482091b760b784f8f1e2f787b6
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/91/520c830f300affb486d333542c5f6c54893673085f6341333fd1957a45ade7
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a4/ffc512be57913428f753539fa25707cf58131a818cdbc8daa9b897cdea9187
.pes/objects/a5/309708f7804f8207340240d6f9356c2b4dfb81c8fd463660ee863c536fb0d8
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

```
vboxuser@OSP:~/banana2/PES1UG24CS523-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Q5.1: A branch is a file in .pes/refs/heads/ storing a commit hash, so pes checkout <branch> updates .pes/HEAD to point to that branch reference. Then the working directory is rebuilt from the target commit's tree by creating, deleting, and overwriting files to match that snapshot. The

index must also align with this state. The operation is complex because it risks overwriting uncommitted user changes and must maintain consistency during file system updates.

Q5.2: Dirty state is detected by comparing working directory files with the index using metadata like size and modification time. Then each index entry is compared with the target branch's commit tree using object hashes. If a file differs from the index and also differs in the target branch, it is a conflict. In that case, checkout is refused to prevent overwriting uncommitted changes.

Q5.3: In detached HEAD, commits are created normally and HEAD moves to them, but no branch pointer is updated. Once you switch away, those commits become unreachable. They remain in the object store until garbage collected. Recovery is done by creating a new branch pointing to the latest commit hash.

Q6.1: Garbage collection uses mark-and-sweep: start from all branch heads and HEAD, traverse commits, trees, and blobs, and mark reachable objects using a hash set. Then delete all unmarked objects from storage. With 100k commits and 50 branches, only reachable history is visited, not all stored objects. Each reachable object is processed once.

Q6.2: GC running during commit can delete newly created objects before they are referenced, causing race conditions. A commit may try to reference an object that GC already removed. This leads to repository corruption. Git avoids this using locking, atomic updates, and grace periods so recently created or referenced objects are not collected immediately.